

Surname: Initials: St No.

(1) Print on A4 paper. (2) While you are preparing for the lab session, refer to the laboratory manual to complete the preparation sheet in your own handwriting in coloured ink – no black ink or ordinary pencil may be used. (3) If a heading is not applicable to your experiment, leave it open. (4) You have to leave the sheet at the completed experiment in your report book. (5) If you do not hand in your preparation, you will lose two marks. (6) You will earn either 1 or 2 **bonus marks** depending on the quality of your preparation. (7) Copying of someone else's work will be viewed in a serious light.

1. Practical abbreviation and full nameH E A: ~~HEA~~ Heat**2. Goal**

To measure the change in temperature of water as it is heated by a resistor and determine the specific heat capacity of water.

3. Safety precautions

Be careful with the ~~thermometer~~ thermometer. If it breaks it ~~can~~ cut the glass shards can cut skin, the resistor can get hot and burn skin.

4. Basic equations used in the practical

$$Q = q(t) = It \quad E = Vq \text{ (Energy transferred)}$$

(charge as a function of time)

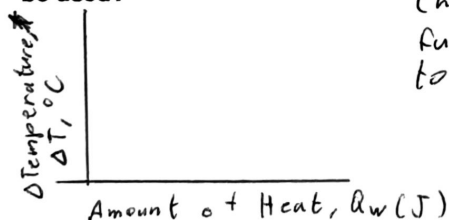
5. Data to be collected

Temperature change of the water, ΔT with respect to the initial temperature, T_0 . The current, I and potential difference, V over the resistor.

6. Table headings including Data collected and calculated data. Mention units of each measurement.

Time, t (min)	Initial Temperature, T_0 ($^{\circ}\text{C}$)	Temperature change, ΔT ($^{\circ}\text{C}$)	Heat, Q_w added (J)	I_{avg} (A), V_{avg} (V)

Table showing change in temperature as the energy added to an amount of water changes with time.

7. Sketch the axes of the graph(s) to be drawn with axis labels and units. What type of graph paper will be used?

Change in temperature of water as a function of the amount of heat added to it.

Normal graph paper will be used

8. What data will be obtained from the graph? Include calculation formula (including units).

The mathematical relationship between the change in temperature and heat added of the water will be obtained.

$$Q_w(\text{J}) = m_w(\text{kg}) c_w(\text{J/kg}^{\circ}\text{C}) \Delta T(^{\circ}\text{C})$$

also the gradient, $m = \frac{\Delta T}{\Delta Q_w} (^{\circ}\text{C/J})$ and y-intercept,